STAINLESS STEEL MANUFACTURING · STEEL MELTING SHOP

AOD Operations

Process Analysis, Standardisation and Business Requirements

Portfolio edition. This is a sanitised version of work carried out in an operations role at a large stainless steel producer. The employer is not named, and all process parameters, quantities, temperatures, gas ranges, equipment identifiers and timings shown are illustrative placeholders, not operating values. The analysis structure, business rules, requirements, controls and validation logic reflect the work as performed. Nothing commercially sensitive is reproduced.

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About This Document

This document consolidates three pieces of process and requirements work carried out in the AOD (Argon Oxygen Decarburisation) area of a Steel Melting Shop. Each part addresses a distinct operational problem, and each follows the same structure: business problem, objectives and scope, stakeholders, current-state analysis, future-state design, rules and controls, and the measures used to confirm the change held.

The AOD converter is the point at which carbon is removed from the melt and the final chemistry of the heat is set. Material additions, gas ratios and temperature windows differ by grade family, and the process runs continuously across three shifts. The recurring problem across all three pieces of work is the same one: operating knowledge sat with experienced individuals rather than in a controlled reference, so execution varied by shift and by operator.

The work was performed in an operations role. The documentation approach, meaning structured requirements, business rules, traceability and acceptance criteria, is the approach used in business analysis, applied here to a manufacturing process rather than a software system.

Document Control

Field	Detail
Prepared by	S. Satyam Singh
Role at the time	Associate Manager, AOD Operations
Business area	Steel Melting Shop, AOD Operations
Coverage	Grade-wise addition pattern standardisation; AOD auto batching requirements; 5S material allocation
Methods applied	Process mapping (As-Is / To-Be), requirements definition, business rules, DMAIC, RACI, control planning, validation scenarios, traceability
Grade families	200 series, 300 series, 400 series
Version	2.0 (portfolio edition)

Contents

Terms Used

Term	Meaning
AOD	Argon Oxygen Decarburisation, the converter process used to refine stainless steel
Main blow / DB1 / DB2	Primary oxygen blowing stage, followed by second and third blow stages
Zero batch	Initial flux charge added before pouring
Dololime / Lime / Flux	Slag-forming additions used to control chemistry and protect refractory
Ferroalloys	Chromium, nickel and manganese bearing additions used to achieve target chemistry
MHS	Material Handling System: conveyors, bunkers, feeders and chutes serving the converter
CH1 / CH2	Charging hoppers feeding material into the converter
Weighing hopper	Intermediate hopper that weighs material before transfer to the charging chute
Reduction	Post-blow stage where chromium is recovered from slag

PART A · Grade-Wise AOD Addition Pattern

Process mapping and standardisation of material addition sequence, process controls and gas switch-over across three grade families.

A1. Background

AOD processing requires controlled addition of slag formers and ferroalloys at defined points in the blowing cycle. Each grade family requires a different addition pattern, flux quantity, alloy sequence, temperature window, stirring duration, reduction condition and gas switch-over point.

Before this work, grade-specific execution was carried out largely from operator experience and verbal handover. The addition sequence was not held in a single structured reference, and there was no consistent basis on which a new Shift-In-Charge could be brought up to competence.

A2. Business Problem

Area	Problem
Grade-wise variation	Each grade family required a different addition pattern, with no single reference covering all three
Process dependency	Operators depended on experience and verbal instruction rather than documented method
Flux control	Flux quantity differed by grade series and required a standard control point
Gas switch-over	Switch-over timing varied by grade and was not consistently documented
Temperature control	Sampling and reduction temperatures needed defined operating ranges
Chemistry achievement	Incorrect addition timing affected achievement of target chemistry
Shift consistency	Each shift required the same reference to produce comparable results
Safety	Enclosure closure and equipment readiness required clear, visible instruction

A3. Objectives

Objective	Description
Standardise addition pattern	Define the correct material addition sequence for each grade series
Improve process consistency	Ensure all shifts follow the same process flow
Improve productivity	Reduce process variation and avoid unnecessary delay in the blowing cycle
Improve chemistry control	Support control of the target element ranges for each grade
Improve gas control	Define switch-over timing by grade
Improve safety compliance	Document the safety points that apply to shop-floor execution
Support training	Enable less experienced team members to work to a defined grade-wise standard

A4. Scope

Area	Included
Grade families	Three series-level addition patterns
Grade-specific controls	Selected grades within each family requiring additional control
Flux addition	Main blow and subsequent blow stage flux addition
Alloy addition	Chromium, nickel and manganese bearing additions
Temperature control	Sampling and reduction temperature ranges
Gas switch-over	Switch-over timing and consumption reference
Safety	Enclosure closure and operational precautions

A5. Stakeholders

Stakeholder	Responsibility
Shift-In-Charge	Controls grade-wise execution during AOD operation
AOD Operator	Follows the defined addition sequence and process instructions
Production Manager	Monitors productivity, heat performance and shift compliance
Quality Team	Verifies chemistry achievement and grade-specific requirements
Safety Team	Ensures safe operation and shop-floor compliance
Maintenance Team	Ensures ladle, launder, chute and porous plug readiness
Gas supply coordination	Supports backup cooling requirement for specific grades

A6. As-Is and To-Be Process View

Area	As-Is	To-Be
Grade-wise process	Grade-specific instructions followed from experience	Standard process flow defined for each series
Addition sequence	Not held in one structured reference	Material addition sequence defined for each grade
Gas switch-over	Timing varied by grade, documented inconsistently	Switch-over condition defined per grade against a controlled range
Flux quantity	Different series required different quantities, applied from memory	Flux limits defined per series and process stage
Safety instruction	Held separately from operating practice	Safety points included in the operating reference
Training	New team members depended on senior guidance	Same reference available to all shifts and to new starters

A7. Process Maps

The maps cover the grade-wise flow from zero batch and pouring through main blow, flux addition, sampling, stirring and reduction. Each map covers one grade family and shows the operational sequence together with its process control window.

Map ID	Grade family	Trigger	Output
PM-01	300 series	Heat ready for AOD processing	Heat ready for reduction stage
PM-02	400 series	Heat ready for AOD processing	Heat ready for reduction stage
PM-03	200 series	Heat ready for AOD processing	Heat ready for reduction stage

A8. Standard Addition Pattern (Representative Sequence)

Sequence structure is as documented. Quantities and temperatures below are illustrative placeholders.

Step	Activity
1	Add zero batch at the series-defined flux quantity
2	Start pouring
3	Start process
4	Add slag former at the standard quantity for the series
5	Add chromium bearing alloy
6	Add flux at the series-defined interim quantity
7	Add nickel bearing alloy where the grade family requires it
8	Add metallic addition
9	Add manganese bearing alloy
10	Add flux up to the series maximum for main blow
11	Continue main blow
12	Add remaining flux during the subsequent blow stages
13	Take sample and temperature within the defined sampling window
14	Complete stirring phase for the defined minimum duration
15	Complete reduction within the defined reduction window and minimum duration

Series-level differences: the 200 series carries a higher interim and maximum flux quantity than the 300 and 400 series; the 300 series includes nickel bearing additions that the 400 series does not; the 400 and 200 series carry a defined minimum reduction duration.

A9. Grade-Specific Process Controls

Individual grades required additional controls beyond the family-level pattern. These were documented to reduce process variation on the grades most sensitive to execution.

Grade type	Key control points
Low-carbon austenitic	Chromium alloy added slowly ahead of the defined oxygen requirement. Flux not to be added while alloy addition is ongoing. Defined sampling and tapping temperature ceilings.
High-nickel austenitic	Backup cooling started before the heat. Tight upper limit on dissolved gas. Extra flux restricted. Ladle life, launder condition, slide gate plate, purging and slag control monitored.
Ferritic	Backup cooling arranged before the heat. Feed material availability confirmed. Dome and mouth cleaning completed before the heat. Both porous plugs confirmed working.
Low-nickel austenitic	Low build-up and controlled tapping sulphur. Opening phosphorus within range. Report checking and additive availability confirmed. Excess flux restricted.

A10. Gas Switch-over Control

Switch-over timing was defined grade by grade to control gas pick-up and improve process stability. At family level the pattern is as follows.

Grade family	Switch-over control
300 series	Timing varies by grade: ahead of the vacuum stage, during reduction, or in the second blow
400 series	Timing defined during main blow or reduction, depending on grade requirement
200 series	Timing linked to percentage completion of the reduction stage

A grade-level reference table was maintained pairing each grade with its controlled gas range and its switch-over trigger, expressed either as a percentage of main blow, a percentage of reduction, or a fixed offset from a process event. Actual ranges and triggers are omitted from this edition.

A11. Business Rules

Rule ID	Business rule
BR-A01	Each stainless steel series must follow its own defined AOD addition pattern
BR-A02	Zero batch must be added before pouring
BR-A03	Slag former must be added after process start, at the standard quantity
BR-A04	Flux addition must follow grade-wise quantity limits
BR-A05	Each series has a defined maximum flux quantity for the main blow stage
BR-A06	Remaining flux must be added during the subsequent blow stages
BR-A07	Sampling temperature must be maintained within the defined window
BR-A08	Reduction temperature must be maintained within the defined window
BR-A09	Stirring phase must meet the defined minimum duration
BR-A10	Where applicable to the series, reduction must meet the defined minimum duration
BR-A11	Gas switch-over must follow the grade-specific requirement
BR-A12	The converter enclosure must remain closed until tapping is complete
BR-A13	Excess flux must not be added for grades where it is restricted
BR-A14	Flux must not be added while alloy addition is ongoing
BR-A15	Charging hopper selection must be fixed before starting any grade sequence
BR-A16	Where opening silicon is high, flux quantity must be increased to process requirement

A12. Safety Requirements

Area	Control
Converter enclosure	Must remain closed until tapping is complete
Ladle condition	Ladle and launder must be clean before critical grades
Porous plug	Both porous plugs must be working where required
Chute condition	Reduction mixture chute must be clean
Purging	Purging must be sufficient to show bubbles at the surface
Mouth and dome	Dome and mouth cleaning completed before the heat where required
Personal protective equipment	Helmet, safety shoes, nose mask, hand gloves and jacket required for shop-floor execution

A13. Measures

Measure	Before standardisation	After standardisation
AOD process time	Variable between shifts	Controlled and consistent
Addition sequence	Experience-based	Reference-driven
Flux and alloy control	Risk of overlapping addition	Clearly separated by rule
Shift compliance	Dependent on individual practice	Measured against documented standard
Operator guidance	Verbal and experience-based	Documented step-by-step instruction
Process variation	Higher chance of variation	Reduced variation